

Madhav K

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EDUCATION

Vellore Institute of Technology, Chennai

Expected May 2027

B.Tech, Computer Science and Engineering (AI and Machine Learning Specialisation) | CGPA: 7.92 / 10

EXPERIENCE

AI Developer | ADMRLS, Chennai

Jul 2026 – Present

- Own end-to-end delivery of AI features for external clients, from requirement scoping through deployment.
- Translate loosely defined client requirements into shippable solutions, working directly with non-technical stakeholders.

Backend and Data Engineering Intern | Tech Mahindra, Chennai

May 2026 – Jul 2026

- Designed PostgreSQL schemas with SQLAlchemy ORM; built modular Pandas ETL pipelines and deployed FastAPI microservices with JWT auth, full CRUD and Docker containerisation.
- Automated a GitHub Actions CI/CD pipeline running PyTest, Docker build and AWS EC2 deploy with a test-first workflow enforced before merge.

Technology Intern | Virtusa, Chennai

May 2025 – Aug 2025

- Supported production NLP pipelines for automated document extraction and end-to-end workflows.
- Handled ML data preparation including preprocessing, annotation and quality checks in Python; wrote utility scripts for cleaning and transformation.

Interdisciplinary Technical Content Intern | FlyBrand Digital Solutions, Chennai

Jun 2025 – Jul 2025

- Built predictive and risk avoidance models on sensor, failure and maintenance log data across Mechanical, Civil and ECE domains.
- Produced technical content explaining AI tools and platforms, translating complex model outputs for non-technical audiences.

PUBLICATION

A Comprehensive Analysis of Lightweight CNN for Fall Detection in Hallway Monitoring. *10th IEEE International Conference on Inventive Systems and Control (ICISC 2026)*, Coimbatore, August 2026.

PROJECTS

EpiGraph-AI: Epidemic Outbreak Prediction | Python, PyTorch, BioBERT

[GitHub](#)

- Designed a hybrid GAT, LSTM and BioBERT architecture predicting dengue outbreaks from spatio-temporal clinical, weather and news data; fused 768-dim BioBERT embeddings reduced via PCA into a multimodal pipeline.
- Evaluated on MAE, RMSE and Outbreak Detection Accuracy using graph adjacency matrices with 7-day rolling windows.

PAD Detection: Predictive Modelling for Peripheral Artery Disease | Python, Scikit-learn

[GitHub](#)

- Engineered 13 clinical features from EHR data; benchmarked 6 classifiers (LightGBM, XGBoost, Random Forest, SVM, LSTM); best AUC 0.88. Hardened against temporal data leakage with class-balanced validation.

Finstox: Explainable AI Stock Trading Platform | FastAPI, PyTorch, LIME, SHAP

[GitHub](#)

- Built a FastAPI backend serving an LSTM model for NSE stock price prediction; added LIME and SHAP per-prediction explanations so non-technical users could interrogate model output.

SKILLS

ML, AI and GenAI: PyTorch, Scikit-learn, TensorFlow, LightGBM, XGBoost, LSTM, GAT, NLP, HuggingFace Transformers, BioBERT, LangChain, RAG, LIME, SHAP, EDA

Backend, Data and DevOps: Python, SQL, FastAPI, Flask, REST APIs, PostgreSQL, SQLAlchemy, ETL, Docker, GitHub Actions CI/CD, AWS (EC2, S3), PyTest, Pandas, NumPy, Seaborn, Java, C++, React.js, Git